

Network Security Implementation

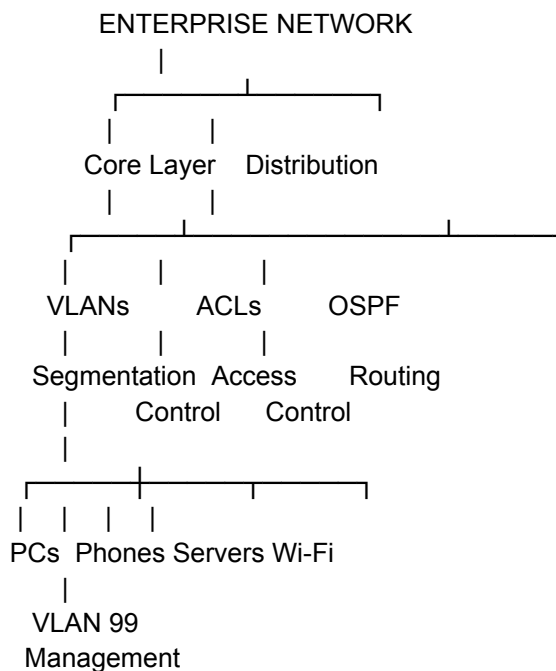
Overview

Security controls were implemented throughout the enterprise network to provide traffic segmentation, controlled access between network segments, secure network management, and protection of infrastructure resources.

The security design combines VLAN segmentation, Access Control Lists (ACLs), secure management practices, port security, and network address translation.

Security Architecture

The lab uses multiple layers of network security rather than relying on a single security control.



These controls work together to separate traffic and restrict unauthorized communication.

1. VLAN Segmentation

VLANs provide logical separation between different types of devices.

The network uses:

VLAN	Purpose
10	PCs
20	Phones
30	Servers
40	Wi-Fi
99	Management

The A-side network uses VLAN 40 for Wi-Fi, while the B-side network uses VLAN 30 for servers.

VLAN segmentation limits unnecessary Layer 2 communication and provides separate network segments that can be controlled using Layer 3 security policies.

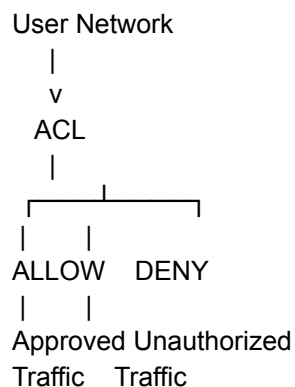
2. Access Control Lists

Access Control Lists (ACLs) are used to control traffic between network segments.

ACLs allow network administrators to define which traffic should be permitted or denied based on criteria such as:

- Source IP address
- Destination IP address
- Protocol
- Port number
- Traffic direction

A simplified ACL policy can be represented as:



ACLs provide an additional security layer between VLANs and routed network segments.

3. Management Network

VLAN 99 is designated as the management network.

Management traffic is separated from normal user traffic so that network infrastructure administration does not share the same logical network as end-user devices.

Management interfaces include Layer 3 SVI addresses on the distribution switches.

Examples include:

A-Side Management:
10.0.0.0/28

B-Side Management:
10.0.0.16/28

Separating management traffic reduces exposure of infrastructure management interfaces to normal user networks.

4. Secure Device Management

Network infrastructure should be administered using secure management protocols whenever supported.

Recommended secure management practices include:

- SSH instead of Telnet
- Strong authentication credentials
- Privileged EXEC authentication
- Encrypted management sessions
- Restricting management access to authorized networks
- Disabling unused services
- Regularly reviewing device configurations

Example secure remote-management configuration:

```
ip domain-name lab.local
crypto key generate rsa
ip ssh version 2
```

SSH provides encrypted communication between an administrator and the network device.

5. Port Security

Access-layer switch ports can be protected using port-security features.

Port security can restrict which MAC addresses are allowed to communicate through an access port.

Example configuration:

```
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
```

This can help prevent unauthorized devices from being connected to protected access ports.

6. DHCP Security

DHCP provides automatic IP addressing to clients.

DHCP-related security controls can help prevent unauthorized devices from providing incorrect DHCP information.

Recommended controls include:

- DHCP snooping
- Trusted DHCP uplinks
- Untrusted access ports
- Dynamic ARP Inspection where supported
- IP Source Guard where appropriate

Example DHCP snooping configuration:

```
ip dhcp snooping
ip dhcp snooping vlan 10,20,30,40
```

Trusted interfaces should be explicitly configured where DHCP server traffic is expected.

7. Network Address Translation

Network Address Translation (NAT) can be used to translate private internal addresses when communicating with external networks.

The internal network uses private IPv4 addressing while R1 provides connectivity toward the external/WAN network.

The external network uses:

203.0.113.4/30

NAT can provide address translation between internal private networks and external addressing.

NAT should not be considered a replacement for firewall or ACL security controls.

8. Default Route Security

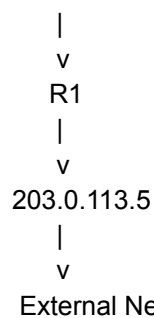
R1 provides a path toward the external network using a static default route.

The upstream next-hop address is:

203.0.113.5

The default route can be represented as:

Internal Network



Restricting and monitoring traffic leaving the internal network is an important component of network security.

9. OSPF Security Considerations

OSPF provides dynamic routing throughout the internal network.

Because routing protocols influence how traffic moves through the network, OSPF should be protected against unauthorized routing peers and route manipulation.

Recommended OSPF security practices include:

- OSPF authentication
- Passive interfaces where neighbor relationships are unnecessary
- Explicitly defining OSPF-enabled interfaces
- Restricting unauthorized devices from participating in routing
- Monitoring OSPF neighbor relationships

Example passive-interface configuration:

```
router ospf 1
passive-interface default
no passive-interface <trusted-interface>
```

This prevents unnecessary OSPF neighbor formation on interfaces that should not establish routing adjacencies.

10. Layered Security Model

The lab demonstrates a defense-in-depth approach.

Layer 1
Physical Security
↓
Layer 2
VLAN Segmentation
↓
Layer 3
ACLs / Routing Controls
↓
Management
SSH / Authentication
↓
Infrastructure
Port Security / DHCP Security
↓
External Connectivity
NAT / Default Route

Using multiple security controls reduces reliance on a single defensive mechanism.

Security Verification

The following commands can be used to verify the security configuration.

Verify ACLs

```
show access-lists
```

This displays configured ACL entries and matching traffic counters.

Verify VLANs

```
show vlan brief
```

This verifies VLAN IDs, names, and active status.

Verify Port Security

```
show port-security
```

This displays port-security status and configuration.

Verify DHCP Snooping

```
show ip dhcp snooping
```

This verifies DHCP snooping configuration and enabled VLANs.

Verify SSH

```
show ip ssh
```

This verifies SSH configuration and version information.

Verify OSPF Neighbors

```
show ip ospf neighbor
```

This verifies established OSPF adjacencies.

Verify Routing

show ip route

This verifies connected, static, and dynamically learned routes.

Security Testing

Security controls should be validated through controlled testing.

ACL Testing

Attempt authorized and unauthorized communication between VLANs.

Expected behavior:

Authorized traffic → PERMITTED

Unauthorized traffic → DENIED

VLAN Testing

Verify that devices are assigned to their intended VLANs.

show vlan brief

Routing Testing

Verify that only expected networks are reachable.

ping <destination>

tracert <destination>

Management Testing

Verify that network management access is available only from authorized management networks.

Security Design Summary

The enterprise network incorporates several security concepts commonly used in production environments:

- VLAN segmentation
- Management network isolation
- ACL-based traffic control
- Secure device management
- Port security
- DHCP security
- OSPF security
- NAT
- Controlled external connectivity
- Defense in depth

These controls demonstrate practical understanding of network security principles and provide a foundation for transitioning from traditional network security into cloud security.

Security+ Alignment

The security concepts demonstrated in this project directly relate to foundational cybersecurity topics covered by CompTIA Security+.

Project Implementation	Security Concept
VLANs	Network Segmentation
ACLs	Access Control
VLAN 99	Management Plane Security
SSH	Secure Administration
Port Security	Network Access Control
DHCP Snooping	Network Attack Mitigation
OSPF Security	Secure Routing
NAT	Network Address Translation
Defense in Depth	Layered Security

This project demonstrates how security controls can be implemented directly within a network infrastructure environment rather than being limited to theoretical knowledge.